The empirical recurrence for OEIS A227638, derived from the entry's generating function

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Abstract

OEIS A227638 carries an empirical recurrence of order 10. The entry separately records a generating function, not as a conjecture but as a statement of fact, and the recurrence is a consequence of it: multiplying the generating function by the polynomial the recurrence supplies gives a POLYNOMIAL, and the recurrence then holds at every index past that polynomial's degree. No model of the objects being counted is needed, and no search: the whole argument is one polynomial division, carried out in exact arithmetic. The result is conditional on the recorded generating function, which is what the entry asserts and what is checked here against every published term.

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1 The two statements

OEIS A227638 is "Number of n X 3 0,1 arrays indicating 2 X 2 subblocks of some larger (n+1) X 4 binary array having determinant equal to one, with rows and columns of the latter in nondecreasing lexicographic order.". It has offset 1 and begins

$$3, 11, 39, 127, 377, 1014, 2518, 5844, \ \ldots$$

The entry states, as a conjecture and never marked settled:

> a(n) = 10*a(n-1) - 45*a(n-2) + 120*a(n-3) - 210*a(n-4) + 252*a(n-5) - 210*a(n-6) + 120*a(n-7) - 45*a(n-8) + 10*a(n-9) - a(n-10) for n>13.

and, separately and NOT as a conjecture:

> G.f.: x*(3 - 19*x + 64*x^2 - 128*x^3 + 172*x^4 - 167*x^5 + 151*x^6 - 154*x^7 + 151*x^8 - 107*x^9 + 49*x^10 - 13*x^11 + 2*x^12) / (1 - x)^10.

As of the "Last modified" line on the live entry (Sep 09 2018 09:23:40, revision 7) the first is still recorded as empirical. This note derives it from the second.

2 A rational generating function decides every linear recurrence

Write

$$G(x) \;=\; \sum_n a(n)\,x^n \;=\; \frac{2x^{13}-13x^{12}+49x^{11}-107x^{10}+151x^9-154x^8+151x^7-167x^6+172x^5-128x^4+6}{x^{10}-10x^9+45x^8-120x^7+210x^6-252x^5+210x^4-120x^3+45x^2-10}$$

the entry's own generating function, and let

$$D(x) \;=\; 1-\sum_i c_i x^i \;=\; x^{10}-10x^9+45x^8-120x^7+210x^6-252x^5+210x^4-120x^3+45x^2-10x+1$$

be the polynomial attached to the conjectured recurrence.

Lemma 1. *For every n in range, $a(n) - \sum_i c_i\, a(n-i)$ is the coefficient of x^n in $D(x)G(x)$.*

Proof. Multiplying power series, the coefficient of x^m in $D(x)G(x)$ is $a_m - \sum_i c_i a_{m-i}$ where a_m denotes the coefficient of x^m in G, terms with negative index being zero. □

Theorem 2.

$$a(n) \;=\; 10\,a(n{-}1){-}45\,a(n{-}2){+}120\,a(n{-}3){-}210\,a(n{-}4){+}252\,a(n{-}5){-}210\,a(n{-}6){+}120\,a(n{-}7){-}45\,a(n{-}8){+}1$$

for every $n > 13$.

Proof. In exact arithmetic,

$$D(x)\,G(x) \;=\; 2x^{13}{-}13x^{12}{+}49x^{11}{-}107x^{10}{+}151x^{9}{-}154x^{8}{+}151x^{7}{-}167x^{6}{+}172x^{5}{-}128x^{4}{+}64x^{3}{-}19x^{2}{+}3x,$$

a POLYNOMIAL of degree 13 — the denominator of G divides $D(x)$ times its numerator, which is precisely the condition for the conjectured recurrence to be implied by the generating function. Its coefficients vanish beyond degree 13, so by Lemma 1 the residual $a(n){-}\sum_i c_i a(n{-}i)$ vanishes for every $n > 13$. The bound is exact: at $n = 13$ the recurrence fails on the entry's own published terms. □

3 Is the conjectured recurrence the shortest one?

Cancelling common factors, G has reduced denominator

$$x^{10} - 10x^9 + 45x^8 - 120x^7 + 210x^6 - 252x^5 + 210x^4 - 120x^3 + 45x^2 - 10x + 1,$$

of degree 10, and that degree is the order of the SHORTEST linear recurrence with constant coefficients the sequence obeys: any shorter one would give a polynomial multiple of G with a smaller denominator, contradicting the cancellation. The conjectured recurrence has order 10.

So the conjecture is the minimal recurrence.

4 The residuals, explicitly

Lemma 1 says the residual $a(n) - \sum_i c_i a(n-i)$ is a coefficient of the polynomial $D(x)G(x)$, so it can be read off the entry's own published terms with no generating function involved at all. Doing that:

index	residual
$n = 11$	49
$n = 12$	-13
$n = 13$	2
$n = 14$	0
$n = 15$	0
$n = 16$	0
$n = 17$	0

The residuals are exactly the coefficients of the polynomial displayed above, in order, and they stop at index 13 because that polynomial has degree 13. A reader who trusts neither the generating function nor the algebra can check the table against the entry's DATA by hand; what the generating function adds is the guarantee that the pattern continues, which no amount of checking terms could establish on its own.

5 Verification

Three checks.

First, the generating function was expanded as a power series in exact rational arithmetic and compared against all 30 terms the entry publishes: it reproduces every one. This is what ties the recorded generating function to the sequence the recurrence is about.

Second, the polynomial division above was carried out symbolically, not numerically, so the statement that $D(x)G(x)$ is a polynomial is exact rather than approximate.

Third, the recurrence was evaluated directly on the series coefficients for sixty consecutive indices beyond $n = 13$ and holds at every one, and at $n = 13$ itself it fails — the degree bound is attained, so the range stated is the true one and not merely a safe one.

Remark 3. The argument is conditional on the generating function, which the entry records as a fact and attributes to a contributor rather than offering as a conjecture. What is shown here is that the empirical recurrence is not an independent guess: it is equivalent to information the entry already contains.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A227638.

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